

+Assignment 1 IPv4 Addressing Foundations

Networking Fundamentals

Assessment	Individual assignment	Total	50 marks
Suggested time	60 minutes	Due date	As posted in Moodle
Submission	One PDF or DOCX	Topic	Classful IPv4 addressing
Filename	Surname_Firstname_A1	Attempts	Follow Moodle settings

Purpose

Complete this assignment to show that you can recognize IPv4 address classes, apply default subnet masks, identify network and broadcast addresses, and check whether a default gateway belongs to the correct network.

Instructions

- Answer every question and show your working where a calculation is required.
- Use the classful A, B, and C rules taught in the lesson. Do not apply custom CIDR or VLSM subnetting in this assignment.
- Network and broadcast addresses cannot be assigned to devices.
- You may use a calculator, but your submitted answers must be your own work.
- Type your answers in this document or write clearly on a printed copy.
- Submit one PDF or DOCX file through the Moodle assignment link by the posted deadline.

Student Details

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Class group	Fellowship in Cyberseucrity and Digital Forensics	Submission date	23/09/2026

Mark Allocation

Section	What is assessed	Marks
Part A	Core concepts	10
Part B	Classful address identification	15
Part C	Network and broadcast analysis	15
Part D	Applied troubleshooting	10
	Total	50

Part A Core Concepts

Answer each question in one to four sentences. Each question is worth 2 marks.

- 1** Define an IPv4 address and explain why the address must be unique on the same network. **[2 marks]**

An IPv4 address is a unique numerical label assigned to each device participating in a network. It must be unique so that each device can be individually identified on the network; two devices using the same IP address would prevent unique identification.

- 2** State one difference between flat and hierarchical addressing. Give one example of each type. **[2 marks]**

A flat address is used mainly within an intranetwork, while a hierarchical address is used across internetworks.

- 3** Compare static and dynamic IP addressing. Name the protocol normally used to assign an address dynamically. **[2 marks]**

A static IP address is manually assigned by an administrator and normally remains constant. A dynamic IP address is assigned automatically when the device starts and requires less user intervention or administration

The protocol normally used for dynamic IP assignment is DHCP (Dynamic Host Configuration Protocol)

- 4** State one difference between classful and classless addressing. What does CIDR stand for? **[2 marks]**

In classful, all networks are the same size but in classless the networks can be different sizes.

- 5** List the four IPv4 configuration items presented in the lesson for normal name-based Internet access. **[2 marks]**

IP address
Subnet Mask
Default Gateway
DNS address

Part B Classful Address Identification

For each address, identify the class, write the network and host pattern, and give the default subnet mask. Each row is worth 3 marks.

IPv4 address	Class	Network and host pattern	Default subnet mask
23.14.6.9	A	23.0.0.0	255.0.0.0
145.80.12.200	B	145.80.0.0	255.255.0.0
198.51.100.27	C	198.51.100.0	255.255.255.0
9.200.10.1	A	9.0.0.0	255.0.0.0
188.44.9.72	B	188.44.0.0	255.255.0.0

Part C Network and Broadcast Analysis

Use the default classful mask for each address. Complete all five requested values. Each address is worth 5 marks.

1 34.72.6.19

Network address	34.0.0.0	Broadcast address	34.255.255.255
First usable host	34.0.0.1	Last usable host	34.255.255.254
Maximum usable hosts	16777214		

2 150.20.7.200

Network address	150.20.0.0	Broadcast address	150.20.255.255
First usable host	150.20.0.1	Last usable host	150.20.255.254
Maximum usable hosts	65534		

3 203.0.113.44

Network address	203.0.113.0	Broadcast address	203.0.113.255
First usable host	203.0.113.1	Last usable host	203.0.113.254
Maximum usable hosts	254		

Part D Applied Troubleshooting

Read the scenario, then answer all five questions.

A workstation in a training room has the following IPv4 settings. The workstation can communicate with some local devices but cannot reach resources on another network.

Setting	Configured value
IPv4 address	201.110.213.28
Subnet mask	255.255.255.0
Default gateway	201.110.214.1
DNS server	8.8.8.8

- 1 Give the network address and broadcast address for the workstation. [2 marks]

Network address: 201.110.213.0
Broadcast address: 201.110.213.255

- 2 Is the configured default gateway valid for this workstation? Explain your answer. [2 marks]

The configured default gateway is not valid for this workstation because the default network address is not in the same network with the workstation so there can't communicate between them.

- 3 Give one valid default gateway address. Explain why the network and broadcast addresses cannot be used as the gateway. [2 marks]

I can give 201.110.213.1 because it's in the workstation network so every host can communicate with the default gateway.

- 4 With the incorrect gateway, what communication may still work and what communication is likely to fail? [2 marks]

Communication with devices on the same local network may still work, but communication with devices on another network or the Internet is likely to fail, because the default gateway is the connection used to reach outside networks and its IP address must be part of the local network.

- 5 A lab contains 25 student computers. Recommend static or dynamic addressing for the computers and name the service that would provide the addresses. [2 marks]

I recommend dynamic IP addressing for the 25 student computers because it requires less human intervention and less administration. The protocol used to provide the addresses is DHCP (Dynamic Host Configuration Protocol).